



中国认可  
国际互认  
检测  
TESTING  
CNAS L0779

# TEST REPORT

**N<sub>o</sub>WJ241103**

**Name of product: Single Phase Smart Energy Meter**

**Type and current range: CA168-M 230V 5(80)A 50Hz**

**Class: Active 1**

**Manufacturer: ShenZhen Calinmeter Co.,ltd.**

**Floor 6,Block A,Qiaode Tech Park,No.7 Guangming  
High-tech Zone,Tianliao Community,Yutang Sub-distri  
ct,Guangming District,Shenzhen**

**Type of test:Proxy test**

**Harbin Research Institute of Electrical  
Instruments Co.,Ltd**



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**SICEM**

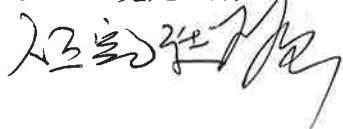
## TEST REPORT

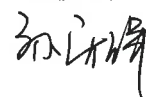
|                             |                                                                                                                                                                                                                                                                                |                 |                                                       |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------|
| Name of product             | Single Phase Smart Energy Meter                                                                                                                                                                                                                                                |                 |                                                       |
| Type and current range      | CA168-M 230V 5(80)A 50Hz                                                                                                                                                                                                                                                       | Class of sample | Active 1                                              |
| Entrust unit                | ShenZhen Calinmeter Co.,Ltd.<br>Floor 6,Block A,Qiaode Tech Park,No.7 Guangming High-tech Zone,Tianliao Community,<br>Yutang Sub-district,Guangming District,Shenzhen<br>Tel: 0755-23707749                                                                                    |                 |                                                       |
| Manufacturer                | ShenZhen Calinmeter Co.,Ltd.<br>Floor 6,Block A,Qiaode Tech Park,No.7 Guangming High-tech Zone,Tianliao Community,<br>Yutang Sub-district,Guangming District,Shenzhen<br>Tel: 0755-23707749                                                                                    |                 |                                                       |
| Number of samples and state | 3<br>normal                                                                                                                                                                                                                                                                    | Sample No.      | №1: 47004042983<br>№2: 47004043007<br>№3: 47004043015 |
| Sample arrival date         | Nov.14,2024                                                                                                                                                                                                                                                                    | Test time       | Nov.14,2024 to Jan.14,2025                            |
| Test type                   | Proxy Test                                                                                                                                                                                                                                                                     | Test item       | 46                                                    |
| Address of the test         | No. 2000 Chuangxin Rd., Songbei District, Harbin, China                                                                                                                                                                                                                        |                 |                                                       |
| Test standard               | IEC62053-21:2020 Electricity metering equipment-Particular requirements-<br>Part 21:Static meters for AC active energy(classes 0.5,1and 2)<br>IEC62052-11:2020 Electricity metering equipment – General requirements, tests and test<br>conditions-Part 11: Metering equipment |                 |                                                       |
| Judge standard              | /                                                                                                                                                                                                                                                                              |                 |                                                       |
| Test conclusion             | The Single Phase Smart Energy Meter meets the requirements of<br>IEC62053-21:2020 and IEC62052-11:2020.<br><br><div style="text-align: right;">           Date: 2025.1.14<br/>           Total pages: 25         </div>                                                        |                 |                                                       |
| Note                        |                                                                                                                                                                                                                                                                                |                 |                                                       |

Test Persons: 丛宪廷 孙久强

Verifier: 王瑜

Ratify: 孙庆辉





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**Test condition:**
 Ambient temperature: (21~24)°C    Relative humidity: (40~60) %



## 1. Test of safety requirements

### 1.1 Tests related to safety

#### 1.1.1 AC voltage test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: The test voltage shall be substantially sinusoidal, having a frequency between 45Hz and 65Hz, and applied for 1 min. During this test no flashover, disruptive discharge or puncture shall occur.
3. Test equipment: Tester 860A (4540688)
4. Test result:

| Test Voltage<br>r.m.s          | Test Result |             |             |
|--------------------------------|-------------|-------------|-------------|
|                                | №1          | №2          | №3          |
| Between circuit and earth: 3kV | Pass        | Pass        | Pass        |
| Test conclusion                | <b>Pass</b> | <b>Pass</b> | <b>Pass</b> |

#### 1.1.2 Impulse voltage test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: The impulse voltage is applied ten times with one polarity and then repeated with the other polarity. The minimum time between the impulses shall be 3s.
3. Test equipment: Tester XTS-11A (03052)
4. Test result:

| Impulse Voltage                    | Test result |             |             |
|------------------------------------|-------------|-------------|-------------|
|                                    | №1          | №2          | №3          |
| Impulse waveform 1.2/50μs, 6.4kV   | Pass        | Pass        | Pass        |
| $\cos\Phi=1.0$ , $I_{n0}$ error(%) | 0.06        | 0.10        | 0.12        |
| Test conclusion                    | <b>Pass</b> | <b>Pass</b> | <b>Pass</b> |

#### 1.1.3 Temperature limits and resistance to heat

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: With each current circuit of the meter carrying maximum current and with each voltage circuit carrying 1.15 times the reference voltage, the temperature rise of the external surface shall not exceed 85°C, with an ambient temperature of 40°C.
3. Test equipment: Digital thermometer WMY-01(212)
4. Test result:

Power factor:  $\cos\Phi=1.0$  Voltage: 264.5V Reference frequency: 50Hz Current: 80A Temperature: 23°C

| Test condition                                                                                                                                                                |                                                         | Limits of temperature rise<br>(°C) | Test result (K) |             |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------------------------------------|-----------------|-------------|-------------|
|                                                                                                                                                                               |                                                         |                                    | №1              | №2          | №3          |
| With each current circuit of the meter carrying maximum current and with each voltage circuit carrying 1.15 times the reference voltage, with an ambient temperature of 23°C. | Surface temperature limits for protection against burns | 100                                | 72              | 73          | 72          |
|                                                                                                                                                                               | Temperature limits for terminals                        | 120                                | 85              | 86          | 86          |
| Test conclusion                                                                                                                                                               |                                                         |                                    | <b>Pass</b>     | <b>Pass</b> | <b>Pass</b> |

### 1.1.4 Spring hammer test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: The meter shall be mounted in its normal working position and the spring hammer shall act on the outer surfaces of the meter cover (including windows) and on the terminal cover with a kinetic energy of  $0.2J \pm 0.02J$ .
3. Test equipment: Impact test bench TY2(027)
4. Test result:

| Test part       | Test requirement | Test result |      |      |
|-----------------|------------------|-------------|------|------|
|                 |                  | №1          | №2   | №3   |
| Outside surface | No damage        | Pass        | Pass | Pass |
| Terminal cover  | No damage        | Pass        | Pass | Pass |
| Test conclusion |                  | Pass        | Pass | Pass |

### 1.1.5 Test of resistance to heat and fire

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: The contact with the glow wire may occur at any random location. If the terminal block is integral with the meter base, it is sufficient to carry out the test only on the terminal block.
3. Test equipment: Hot wire test device ZHZ13 (35093)
4. Test result:

| Location        | Temperature (°C) | Requirement | Test result |      |      |
|-----------------|------------------|-------------|-------------|------|------|
|                 |                  |             | №1          | №2   | №3   |
| Terminal block  | $960 \pm 15$     | No burn     | Pass        | Pass | Pass |
| Terminal Cover  | $650 \pm 10$     | No burn     | Pass        | Pass | Pass |
| Case            | $650 \pm 10$     | No burn     | Pass        | Pass | Pass |
| Test conclusion |                  |             | Pass        | Pass | Pass |

### 1.1.6 Test of protection against penetration of dust

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: IEC62053-21:2020 IEC62052-11:2020
3. Test equipment: Dust test chamber SC-500 (990122)
4. Test result:

| Test condition  | Requirement                                                                                                                                | Test result |      |      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|------|
|                 |                                                                                                                                            | №1          | №2   | №3   |
| IP5X            | Any ingress of dust shall be only in a quantity not impairing the operation of the meter and its dielectric strength (insulating strength) | Pass        | Pass | Pass |
| Test conclusion |                                                                                                                                            | Pass        | Pass | Pass |



### 1.1.7 Test of protection against penetration of water

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: IEC62053-21:2020 IEC62052-11:2020
3. Test equipment: rain test device CNW0115-034WS (THS20212038OS)
4. Test result:

| Test condition  | Requirement                                                                                                                                | Test result |             |             |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|-------------|
|                 |                                                                                                                                            | №1          | №2          | №3          |
| IPX6            | Any ingress of dust shall be only in a quantity not impairing the operation of the meter and its dielectric strength (insulating strength) | Pass        | Pass        | Pass        |
| Test conclusion |                                                                                                                                            | <b>Pass</b> | <b>Pass</b> | <b>Pass</b> |

## 2. Tests of mechanical requirements

### 2.1 Shock test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: The meter shall show no damage or change of the information and shall operate correctly in accordance with the requirements of this standard after the test.
3. Test equipment: Shock test bench CP-100(120807) and Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Current: 5(80)A Reference voltage :230V

| Current         | Power factor<br>(cosΦ) | Percentage error limits<br>(%) | Test result (%) |             |             |
|-----------------|------------------------|--------------------------------|-----------------|-------------|-------------|
|                 |                        |                                | №1              | №2          | №3          |
| $I_n$           | 1.0                    | ±1.0                           | 0.07            | 0.10        | 0.12        |
| Test conclusion |                        |                                | <b>Pass</b>     | <b>Pass</b> | <b>Pass</b> |

### 2.2 Vibration test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: The meter shall show no damage or change of the information and shall operate correctly in accordance with the requirements of this standard after the test.
3. Test equipment: Vibration test bench D-1000-5(920407) and Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Current: 5(80)A Reference voltage :230V

| Current         | Power factor<br>(cosΦ) | Percentage error limits<br>(%) | Test result (%) |             |             |
|-----------------|------------------------|--------------------------------|-----------------|-------------|-------------|
|                 |                        |                                | №1              | №2          | №3          |
| $I_n$           | 1.0                    | ±1.0                           | 0.09            | 0.12        | 0.10        |
| Test conclusion |                        |                                | <b>Pass</b>     | <b>Pass</b> | <b>Pass</b> |

## 2.3 Terminals – Terminal block(s) (No requirement)

## 3. Tests of general requirements

### 3.1 Power consumption

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: The power consumption in the voltage and current circuit at reference values of the influence quantities.
3. Test equipment: Low power factor wattmeter D5-W (20149) and DMM 7150(303645)
4. Test result:

Power factor :  $\cos\Phi=1.0$  Reference frequency: 50Hz Current: 5(80)A Reference voltage: 230V

| Test item       | Specified range of power consumption (active) | Test result (%) |             |             |
|-----------------|-----------------------------------------------|-----------------|-------------|-------------|
|                 |                                               | №1              | №2          | №3          |
| Voltage circuit | 5 W                                           | 0.58W           | 0.60W       | 0.58W       |
|                 | 25VA                                          | 0.92VA          | 0.95VA      | 0.95VA      |
| Current circuit | 1VA                                           | 0.15VA          | 0.14VA      | 0.14VA      |
| Test conclusion |                                               | <b>Pass</b>     | <b>Pass</b> | <b>Pass</b> |

### 3.2 Meter marking

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: It shall be verified that the relation between the test output and the indication on the display complies with the marking on the name-plate.
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

| Test condition                                                                                                                                                                                                                          | Test result |             |             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|-------------|
|                                                                                                                                                                                                                                         | №1          | №2          | №3          |
| A meter shall bear all of the markings required by local regulations. In addition, and if not already required by the local regulations, the meter shall also bear the following information as applicable: IEC62052-11:2020 6.2 a)~ p) | <b>Pass</b> | <b>Pass</b> | <b>Pass</b> |
| Test conclusion                                                                                                                                                                                                                         | <b>Pass</b> | <b>Pass</b> | <b>Pass</b> |

## 4 Tests of accuracy requirements

### 4.1 Meter constant

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: It shall be verified that the relation between the test output and the indication on the display complies with the marking on the name-plate.
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency: 50Hz Current: 80A Reference voltage: 230V

| Power factor<br>(cosΦ) | Test result (imp/kWh) |             |             |
|------------------------|-----------------------|-------------|-------------|
|                        | №1                    | №2          | №3          |
| 1.0                    | 1000                  | 1000        | 1000        |
| Test conclusion        | <b>Pass</b>           | <b>Pass</b> | <b>Pass</b> |

### 4.2 Initial start-up of the meter

1. Requirement: IEC62053-21:2003 IEC62052-11: 2003
2. Test method: IEC62053-21:2003 IEC62052-11: 2003
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency: 50Hz Current: 5(80)A Reference voltage: 230V

| Test item        | Test result |             |             |
|------------------|-------------|-------------|-------------|
|                  | №1          | №2          | №3          |
| Initial start-up | Pass        | Pass        | Pass        |
| Test conclusion  | <b>Pass</b> | <b>Pass</b> | <b>Pass</b> |

### 4.3 Test of no-load condition

1. Requirement: IEC62053-21:2020 IEC62053-23:2020 IEC62052-11:2020
2. Test method: The current circuit shall be open circuit and a voltage of 110% of the reference voltage shall be applied to the voltage circuits.
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

Power factor: cosΦ=1.0 Reference frequency: 50Hz Current: 5(80)A Reference voltage: 230V

| Voltage                       | Test result |             |             |
|-------------------------------|-------------|-------------|-------------|
|                               | №1          | №2          | №3          |
| 110% of the reference voltage | Pass        | Pass        | Pass        |
| Test conclusion               | <b>Pass</b> | <b>Pass</b> | <b>Pass</b> |

#### 4.4 Starting current test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: The meter shall start and continue to register  $0.004I_n$ .
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency: 50Hz Current: 5(80)A Reference voltage: 230V

| Power factor<br>( $\cos\Phi$ ) | Current    | Test Result |      |      |
|--------------------------------|------------|-------------|------|------|
|                                |            | №1          | №2   | №3   |
| 1.0                            | $0.004I_n$ | Pass        | Pass | Pass |
| Test conclusion                |            | Pass        | Pass | Pass |

#### 4.5 Repeatability test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: When the meter is under the reference conditions, the percentage errors shall not exceed the limits.
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency: 50Hz Current: 5(80)A Reference voltage: 230V

| Current         | Power factor<br>( $\cos\Phi$ ) | Percentage error limits<br>(%) | Test result (%) |      |       |
|-----------------|--------------------------------|--------------------------------|-----------------|------|-------|
|                 |                                |                                | №1              | №2   | №3    |
| $I_{min}$       | 1.0                            | $\pm 0.3$                      | 0.03            | 0.04 | 0.02  |
| $0.1I_n$        | 1.0                            | $\pm 0.2$                      | 0.01            | 0.02 | 0.02  |
| $I_n$           | 1.0                            | $\pm 0.2$                      | 0.01            | 0.02 | 0.02  |
| $I_{max}$       | 1.0                            | $\pm 0.2$                      | 0.01            | 0.02 | 0.01  |
| $0.1I_n$        | 0.5L                           | $\pm 0.3$                      | -0.01           | 0.01 | -0.01 |
| $I_n$           | 0.5L                           | $\pm 0.2$                      | -0.02           | 0.01 | 0.02  |
| $I_{max}$       | 0.5L                           | $\pm 0.2$                      | 0.00            | 0.02 | 0.03  |
| $0.1I_n$        | 0.8C                           | $\pm 0.3$                      | 0.03            | 0.02 | 0.02  |
| $I_n$           | 0.8C                           | $\pm 0.2$                      | 0.04            | 0.01 | 0.02  |
| $I_{max}$       | 0.8C                           | $\pm 0.2$                      | 0.02            | 0.04 | 0.01  |
| Test conclusion |                                |                                | Pass            | Pass | Pass  |

#### 4.6 Limits of error due to variation of the current

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: When the meter is under the reference conditions, the percentage errors shall not exceed the limits.
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency: 50Hz Current: 5(80)A Reference voltage: 230V

| Current         | Power factor<br>(cosΦ) | Percentage error limits<br>(%) | Test result (%) |       |       |
|-----------------|------------------------|--------------------------------|-----------------|-------|-------|
|                 |                        |                                | №1              | №2    | №3    |
| $I_{\min}$      | 1.0                    | ±1.5                           | 0.23            | 0.30  | 0.28  |
| $0.1I_n$        | 1.0                    | ±1.0                           | 0.12            | 0.15  | 0.10  |
| $0.5I_n$        | 1.0                    | ±1.0                           | 0.07            | 0.07  | 0.09  |
| $I_n$           | 1.0                    | ±1.0                           | 0.13            | 0.09  | 0.10  |
| $0.5I_{\max}$   | 1.0                    | ±1.0                           | 0.05            | 0.08  | 0.06  |
| $I_{\max}$      | 1.0                    | ±1.0                           | 0.06            | 0.07  | 0.04  |
| $0.1I_n$        | 0.5L                   | ±1.5                           | 0.12            | 0.15  | 0.11  |
| $0.2I_n$        | 0.5L                   | ±1.0                           | 0.07            | 0.10  | 0.03  |
| $0.5I_n$        | 0.5L                   | ±1.0                           | 0.04            | 0.05  | 0.01  |
| $I_n$           | 0.5L                   | ±1.0                           | 0.04            | 0.06  | 0.01  |
| $0.5I_{\max}$   | 0.5L                   | ±1.0                           | 0.03            | 0.01  | -0.06 |
| $I_{\max}$      | 0.5L                   | ±1.0                           | -0.03           | -0.04 | -0.08 |
| $0.1I_n$        | 0.8C                   | ±1.5                           | 0.12            | 0.10  | 0.15  |
| $0.2I_n$        | 0.8C                   | ±1.0                           | 0.07            | 0.12  | 0.11  |
| $0.5I_n$        | 0.8C                   | ±1.0                           | 0.13            | 0.11  | 0.14  |
| $I_n$           | 0.8C                   | ±1.0                           | 0.12            | 0.16  | 0.14  |
| $0.5I_{\max}$   | 0.8C                   | ±1.0                           | 0.07            | 0.02  | 0.11  |
| $I_{\max}$      | 0.8C                   | ±1.0                           | 0.09            | -0.18 | 0.10  |
| Test conclusion |                        |                                | Pass            | Pass  | Pass  |



#### 4.7 Test of time keeping accuracy

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: IEC62053-21:2020 IEC62052-11:2020
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

| Test requirements                                                                                                                               | Test result |      |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|------|
|                                                                                                                                                 | №1          | №2   | №3   |
| At reference voltage and reference temperature,crystal-controlled time switches shall have a time-keeping accuracy better than $\pm 0,5$ s/day. | Pass        | Pass | Pass |
| The variation of the time-keeping accuracy with the temperature shall be less than( $\pm 0,15$ s/C/24h).                                        | Pass        | Pass | Pass |
| On operation reserve,at reference temperature, the time-keeping accuracy shall be better than $\pm 1$ s /day.                                   | Pass        | Pass | Pass |
| Test conclusion                                                                                                                                 | Pass        | Pass | Pass |

### 5 Tests for electromagnetic compatibility (EMC) and limits of error due to influence quantities

#### 5.1 Limits of percentage error due to influence quantities

See details 5.2~5.13

## 5.2 Voltage dips and short interruptions

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: IEC62053-21:2020 IEC62052-11:2020
3. Test equipment: Cycle drop device VDS-1132A(VDS-1132A0130101)and Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Reference voltage :230V

| Interrupt and interruption voltage and time | The function of judgment     | Limits of change in the register | Test result |      |      |
|---------------------------------------------|------------------------------|----------------------------------|-------------|------|------|
|                                             |                              |                                  | №1          | №2   | №3   |
| $\Delta U=100\%$<br>5 cycle                 | Register Function            | $\leq 0.0184kWh$                 | Pass        | Pass | Pass |
|                                             | Output pulse                 | No output                        | Pass        | Pass | Pass |
| $\Delta U=100\%$<br>50 cycle                | Register Function            | $\leq 0.0184kWh$                 | Pass        | Pass | Pass |
|                                             | Output pulse                 | No output                        | Pass        | Pass | Pass |
| $\Delta U=100\%$<br>1 cycle                 | Register Function            | $\leq 0.0184kWh$                 | Pass        | Pass | Pass |
|                                             | Output pulse                 | No output                        | Pass        | Pass | Pass |
| $\Delta U=95\%$<br>250 cycle                | Register Function            | $\leq 0.0184kWh$                 | Pass        | Pass | Pass |
|                                             | Output pulse                 | No output                        | Pass        | Pass | Pass |
| $\Delta U=60\%$<br>5 cycle                  | Register Function            | $\leq 0.0184kWh$                 | Pass        | Pass | Pass |
|                                             | Output pulse                 | No output                        | Pass        | Pass | Pass |
| $\Delta U=60\%$<br>50 cycle                 | Register Function            | $\leq 0.0184kWh$                 | Pass        | Pass | Pass |
|                                             | Output pulse                 | No output                        | Pass        | Pass | Pass |
| $\Delta U=30\%$<br>0.5 cycle                | Register Function            | $\leq 0.0184kWh$                 | Pass        | Pass | Pass |
|                                             | Output pulse                 | No output                        | Pass        | Pass | Pass |
| $\Delta U=30\%$<br>1 cycle                  | Register Function            | $\leq 0.0184kWh$                 | Pass        | Pass | Pass |
|                                             | Output pulse                 | No output                        | Pass        | Pass | Pass |
| $\Delta U=50\%$<br>3000 cycle               | Register Function            | $\leq 0.0184kWh$                 | Pass        | Pass | Pass |
|                                             | Output pulse                 | No output                        | Pass        | Pass | Pass |
| $I_n \cos\Phi=1.0$                          | The error (%) after the test |                                  | 0.11        | 0.09 | 0.12 |
| Test conclusion                             |                              |                                  | Pass        | Pass | Pass |

## 5.3 Electrostatic discharge immunity test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: IEC62053-21:2020
3. Test equipment: Electrostatic discharge generator ESD-203A(ESD-203A0130102)
4. Test result:

Reference frequency :50Hz Reference voltage: 230V

| Test conditions                                                   | The function of judgment     | Limits of change in the register | Test result |      |      |
|-------------------------------------------------------------------|------------------------------|----------------------------------|-------------|------|------|
|                                                                   |                              |                                  | №1          | №2   | №3   |
| Indirect discharge 8kV, contact discharge 8kV, air discharge 15kV | Register Function            | ≤0.0184kWh                       | Pass        | Pass | Pass |
| $I_n \cos\Phi=1.0$                                                | The error (%) after the test |                                  | 0.12        | 0.15 | 0.14 |
| Test conclusion                                                   |                              |                                  | Pass        | Pass | Pass |

### 5.4 Radiated, radio-frequency, electromagnetic field immunity test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: IEC62052-11:2020
3. Test equipment: Signal generator SMB100A (115554)  
 Power amplifier AS0104-200/200 (1034148)  
 Power amplifier AP32MT255 (S/N:0906-943)  
 Broadband horn antenna BBHA9120J (S/N:00180)  
 Log periodic antenna STLP9128E (N#1155)

#### 4. Test result:

Test without current

Frequency: 50Hz Reference voltage: 230V

| Test conditions                                                                                                                                                                                                                      | Criteria B                                                       | Test result |      |      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------|------|------|
|                                                                                                                                                                                                                                      |                                                                  | №1          | №2   | №3   |
| Frequency band: 80 MHz-2 GHz,<br>unmodulated field strength 30 V/m;<br>Frequency band: 2 GHz MHz to 6 GHz,<br>unmodulated field strength of 10 V/m;<br>carrier modulated with 80 % AM,<br>at 1 kHz sine wave;<br>The dwell time: 3 s | Limits of change<br>in the register<br>$\leq 0.0184 \text{ kWh}$ | Pass        | Pass | Pass |
|                                                                                                                                                                                                                                      | indicating display                                               | Pass        | Pass | Pass |
| Test conclusion                                                                                                                                                                                                                      |                                                                  | Pass        | Pass | Pass |

Test with current

Frequency: 50Hz Current: 5A Reference voltage: 230V

| Test conditions                                                                                                                                                                                                                               | Criteria A                                                      | The maximum variation in percentage error |      |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------|------|------|
|                                                                                                                                                                                                                                               |                                                                 | №1                                        | №2   | №3   |
| Frequency band: 80 MHz to 2 GHz,<br>unmodulated field strength of 10 V/m;<br>Frequency band: 2 GHz MHz to 6 GHz,<br>unmodulated field strength of 3 V/m;<br>carrier modulated with 80 % AM at 1 kHz<br>sine wave;<br>The frequency step: 1 %; | Limits of the<br>variation (%):<br>2.0<br>( $\cos \Phi = 1.0$ ) | 0.21                                      | 0.30 | 0.31 |
|                                                                                                                                                                                                                                               | indicating<br>display                                           | Pass                                      | Pass | Pass |
| Test conclusion                                                                                                                                                                                                                               |                                                                 | Pass                                      | Pass | Pass |

### 5.5 Electrical fast transient/burst immunity test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: IEC62052-11:2020
3. Test equipment: Burst generator EFT-406N (EFT0460905)
4. Test result:

Frequency :50Hz Reference voltage :230V Current :5A

| Test conditions                                  | Criteria A         |                                                  | Test result |             |             |
|--------------------------------------------------|--------------------|--------------------------------------------------|-------------|-------------|-------------|
|                                                  |                    |                                                  | №1          | №2          | №3          |
| Test Voltage: 4kV<br>Repetition rate:<br>100 kHz | $I_n \cos\Phi=1.0$ | Acceptable limit of<br>the variation (%):<br>4.0 | 0.13        | 0.21        | 0.15        |
|                                                  | Criteria B         | indicating display                               | Pass        | Pass        | Pass        |
| Test conclusion                                  |                    |                                                  | <b>Pass</b> | <b>Pass</b> | <b>Pass</b> |

### 5.6 Immunity to conducted disturbances, induced by radio-frequency fields

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: IEC62053-21:2020 IEC62052-11:2020
3. Test equipment: NSG4070-75 Conducted disturbances test system (35761)
4. Test result:

Reference voltage: 230V Power factor:  $\cos\Phi=1.0$  Reference frequency :50Hz Current :5(80)A

| Test voltage    | Frequency (MHz) | Limits of variation (%) | Test result |      |      |
|-----------------|-----------------|-------------------------|-------------|------|------|
|                 |                 |                         | №1          | №2   | №3   |
| 10V             | 0.15            | 2.0                     | 0.05        | 0.08 | 0.12 |
|                 | 0.5             |                         | 0.10        | 0.13 | 0.12 |
|                 | 1.0             |                         | 0.05        | 0.12 | 0.10 |
|                 | 2.0             |                         | 0.07        | 0.08 | 0.12 |
|                 | 5.0             |                         | 0.12        | 0.05 | 0.07 |
|                 | 10.0            |                         | 0.10        | 0.05 | 0.08 |
|                 | 20.0            |                         | 0.12        | 0.12 | 0.10 |
|                 | 30.0            |                         | 0.13        | 0.11 | 0.12 |
|                 | 50.0            |                         | 0.08        | 0.02 | 0.13 |
|                 | 80.0            |                         | 0.10        | 0.05 | 0.10 |
| Test conclusion |                 |                         | Pass        | Pass | Pass |

Reference voltage: 230V Reference frequency:50Hz

| Test content                     | Limits of charge<br>in the register | Test result |             |             |
|----------------------------------|-------------------------------------|-------------|-------------|-------------|
|                                  |                                     | №1          | №2          | №3          |
| Frequency range:<br>150kHz~80MHz | $\leq 0.0184\text{kWh}$             | Pass        | Pass        | Pass        |
|                                  | No output                           | Pass        | Pass        | Pass        |
| Test conclusion                  |                                     | <b>Pass</b> | <b>Pass</b> | <b>Pass</b> |

## 5.7 Test for Immunity to conducted, differential mode disturbances and signalling in the frequency range 2 kHz to 150 kHz at AC power ports

- Requirement: IEC62053-21:2020 IEC62052-11:2020
- Test method: IEC62052-11:2020
- Test equipment: Differential-mode conduction harassment simulator IMU-MGS-SMART (109938-1583), IMUSLAVE EXT-SMART II (107128-1501)
- Test result:

Frequency : 50Hz voltage: 230V Current : 5A

| Test conditions                                                                                | Criteria A         |                                               | Test result |             |             |
|------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------|-------------|-------------|-------------|
|                                                                                                |                    |                                               | №1          | №2          | №3          |
| 2 kHz to 30 kHz:<br>I diff= 3 A,<br>30 kHz to 150 kHz:<br>I diff= 1,5 A.<br>Frequency step: 1% | $I_n \cos\Phi=1.0$ | Acceptable limit of the variation (%):<br>4.0 | 0.10        | 0.20        | 0.17        |
|                                                                                                | Criteria B         | indicating display                            | Pass        | Pass        | Pass        |
| Test conclusion                                                                                |                    |                                               | <b>Pass</b> | <b>Pass</b> | <b>Pass</b> |

## 5.8 Surge immunity test

- Requirement: IEC62053-21:2020 IEC62052-11:2020
- Test method: IEC62052-11:2020
- Test equipment: Surge generator LSG-510A (LSG05100905)
- Test result:

Reference frequency : 50Hz Reference voltage: 230V

| Test conditions                                           | The function of judgment            | Requirements                          | Test result |      |      |
|-----------------------------------------------------------|-------------------------------------|---------------------------------------|-------------|------|------|
|                                                           |                                     |                                       | №1          | №2   | №3   |
| Test Voltage: 4kV<br>phase angles:<br>0°, 90°, 180°, 270° | Register Function                   | ≤0.0184kWh                            | Pass        | Pass | Pass |
|                                                           | indicating display                  | permit display<br>quality degradation | Pass        | Pass | Pass |
| $I_n \cos\Phi=1$                                          | The percentage error after the test |                                       | 0.12        | 0.08 | 0.10 |
| Test conclusion                                           |                                     |                                       | Pass        | Pass | Pass |

## 5.9 Ring wave immunity test

- Requirement: IEC62052-11:2020 IEC62053-23:2020
- Test method: IEC62052-11:2020
- Test equipment: Ring wave generator OCS500N (P180621438)
- Test result:

Reference frequency : 50Hz Reference voltage: 220V

| Test conditions                                                                     | The function of judgment            | Requirements                          | Test result |      |      |
|-------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------|-------------|------|------|
|                                                                                     |                                     |                                       | №1          | №2   | №3   |
| line to ground:4kV<br>differential mode:2kV<br>phase angles:<br>0°, 90°, 180°, 270° | Register Function                   | ≤0.0184kWh                            | Pass        | Pass | Pass |
|                                                                                     | indicating display                  | Permit display<br>quality degradation | Pass        | Pass | Pass |
| $I_n \cos\Phi=1$                                                                    | The percentage error after the test |                                       | 0.15        | 0.07 | 0.11 |
| Test conclusion                                                                     |                                     |                                       | Pass        | Pass | Pass |



### 5.10 Damped oscillatory wave immunity test(not applicable)

### 5.11 External static magnetic fields

1. Requirement: IEC62053-21:2003 IEC62052-11:2020
2. Test method: IEC62052-11:2020
3. Test equipment: DC steady current source YJ-10A(6102) and Meter calibration device ST9001D5 (7131019).
4. Test result:

Reference frequency :50Hz Current: 5A Reference voltage :230V

| Test conditions                   | Location           | Power factor (cosΦ) | Limits of variation (%) | Test result (%) |      |      |
|-----------------------------------|--------------------|---------------------|-------------------------|-----------------|------|------|
|                                   |                    |                     |                         | №1              | №2   | №3   |
| the magneto-motive force: 1000 At | Front              | 1.0                 | 2.0                     | 0.09            | 0.09 | 0.11 |
|                                   | Up                 | 1.0                 | 2.0                     | 0.08            | 0.08 | 0.08 |
|                                   | Left               | 1.0                 | 2.0                     | 0.14            | 0.09 | 0.06 |
|                                   | Right              | 1.0                 | 2.0                     | 0.08            | 0.09 | 0.08 |
|                                   | indicating display |                     |                         | Pass            | Pass | Pass |
| Test conclusion                   |                    |                     |                         | Pass            | Pass | Pass |

### 5.12 Power frequency magnetic field immunity test

1. Requirement: IEC62053-21:2003 IEC62052-11:2020
2. Test method: IEC62052-11:2020
3. Test equipment: Meter calibration device ST9001D5 (7131019) and External magnetic field test bench ZHZ26A(09021)
4. Test result:

Reference frequency: 50Hz Current: 5A Reference voltage :230V

| Test conditions                 | Power factor (cosΦ) | Dimension of magnetic | Limits of variation (%) | Test result (%) |      |      |
|---------------------------------|---------------------|-----------------------|-------------------------|-----------------|------|------|
|                                 |                     |                       |                         | №1              | №2   | №3   |
| field strength 0.5 mT (400 A/m) | 1.0                 | X-axis                | 2.0                     | 0.10            | 0.12 | 0.07 |
|                                 |                     |                       | 2.0                     | 0.09            | 0.09 | 0.05 |
|                                 |                     | Y-axis                | 2.0                     | 0.04            | 0.12 | 0.10 |
|                                 |                     |                       | 2.0                     | 0.04            | 0.09 | 0.11 |
|                                 |                     | Z-axis                | 2.0                     | 0.12            | 0.10 | 0.08 |
|                                 |                     |                       | 2.0                     | 0.15            | 0.05 | 0.03 |
|                                 |                     | indicating display    |                         |                 | Pass | Pass |
| Test conclusion                 |                     |                       | Pass                    | Pass            | Pass |      |



### 5.13 Emission requirements

- Requirement: IEC62052-11:2020
- Test method: IEC62052-11:2020
- Test equipment: Electromagnetic disturbance test receiver PMM9010 (696WX21006)  
 Electromagnetic disturbance test receiver PMM9030 (121WX10404)  
 Log periodic antenna LP-02 (0011X51001)  
 Biconical dipole antenna BC-01 (0011X20703)  
 Three-phase power supply network of artificial L3-32 (0120X90108)

#### 4. Test result:

##### a. Limit value of conducted disturbances:

Reference voltage: 230V Power factor: 1.0 Reference frequency: 50Hz Current: 0.5A

| Frequency<br>(MHz) | Limits of quasi peak value<br>dB(μV) | Limits of average value<br>dB(μV) | Test result of average value(μV) |
|--------------------|--------------------------------------|-----------------------------------|----------------------------------|
|                    |                                      |                                   | №1                               |
| 0.15~0.50          | 66~56                                | 56~46                             | ≤31                              |
| 0.50~5             | 56                                   | 46                                | ≤26                              |
| 5~30               | 60                                   | 50                                | ≤28                              |
| Test conclusion    |                                      |                                   | <b>Pass</b>                      |

##### b. Limit value of radiation disturbances:

Reference voltage: 230V Power factor: 1.0 Reference frequency: 50Hz Current: 0.5A

| Frequency<br>(MHz) | Limits of quasi peak value dB(dBμV/m) | Test result of peak value(dBμV/m) |
|--------------------|---------------------------------------|-----------------------------------|
|                    |                                       | №1                                |
| 30~230             | 40                                    | ≤25                               |
| 230~1000           | 47                                    | ≤33                               |
| Test conclusion    |                                       | <b>Pass</b>                       |

## 6 Tests of immunity to other influence quantities

### 6.1 Limits of error due to influence quantities

See details 6.2~6.16

### 6.2 Harmonics in the current and voltage circuits 5th harmonic test

- Requirement: IEC62053-21:2020 IEC62052-11:2020
- Test method: Test the variation in percentage errors at power with harmonic.
- Test equipment: Meter calibration device ST9001D5 (7131019)
- Test result:

Reference frequency: 50Hz Current: 5(80)A Reference voltage: 230V

| Influence quantity                                              | Current      | Power factor<br>(cosΦ) | Limits of variation<br>(%) | Test result (%) |             |             |
|-----------------------------------------------------------------|--------------|------------------------|----------------------------|-----------------|-------------|-------------|
|                                                                 |              |                        |                            | №1              | №2          | №3          |
| Harmonics in the current and voltage circuits 5th harmonic test | $0.5I_{max}$ | 1.0                    | 0.8                        | 0.02            | 0.03        | 0.03        |
| Test conclusion                                                 |              |                        |                            | <b>Pass</b>     | <b>Pass</b> | <b>Pass</b> |

### 6.3 Interharmonics in the current circuit – burst fired waveform test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: Test the variation in percentage errors at power with harmonic.
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Current: 5(80)A Reference voltage: 230V

| Influence quantity                                           | Current  | Power factor (cosΦ) | Limits of variation (%) | Test result (%) |      |      |
|--------------------------------------------------------------|----------|---------------------|-------------------------|-----------------|------|------|
|                                                              |          |                     |                         | №1              | №2   | №3   |
| Interharmonics in the current circuit – burst fired waveform | $0.5I_n$ | 1.0                 | 3.0                     | 0.02            | 0.01 | 0.04 |
| Test conclusion                                              |          |                     |                         | Pass            | Pass | Pass |

### 6.4 Odd harmonics in the current circuit

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: Test the variation in percentage errors at power with harmonic.
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Current: 5(80)A Reference voltage: 230V

| Influence quantity                        | Phase fired waveforms | Current  | Power factor (cosΦ) | Limits of variation (%) | Test result (%) |      |      |
|-------------------------------------------|-----------------------|----------|---------------------|-------------------------|-----------------|------|------|
|                                           |                       |          |                     |                         | №1              | №2   | №3   |
| odd-harmonics in the A.C. Current circuit | 45°                   | $0.5I_n$ | 1.0                 | 3.0                     | 0.14            | 0.16 | 0.14 |
| odd-harmonics in the A.C. Current circuit | 90°                   | $0.5I_n$ | 1.0                 | 3.0                     | 0.12            | 0.10 | 0.12 |
| odd-harmonics in the A.C. Current circuit | 135°                  | $0.5I_n$ | 1.0                 | 3.0                     | 0.11            | 0.15 | 0.13 |
| Test conclusion                           |                       |          |                     |                         | Pass            | Pass | Pass |

### 6.5 DC and even harmonics – half wave rectified waveform test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: Test the variation in percentage errors at power with harmonic.
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Current: 5(80)A Reference voltage: 230V

| Influence quantity                           | Current                    | Power factor (cosΦ) | Limits of variation (%) | Test result (%) |      |      |
|----------------------------------------------|----------------------------|---------------------|-------------------------|-----------------|------|------|
|                                              |                            |                     |                         | №1              | №2   | №3   |
| DC and even harmonics in the current circuit | $\frac{I_{max}}{\sqrt{2}}$ | 1.0                 | 3.0                     | 0.15            | 0.20 | 0.18 |
|                                              |                            | 0.5                 | 3.0                     | 0.23            | 0.25 | 0.24 |
| Test conclusion                              |                            |                     |                         | Pass            | Pass | Pass |

## 6.6 Voltage variation

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: Test the variation in percentage errors when voltage variation.
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Current :5(80)A Reference voltage: 230V

| Current         | Power factor<br>(cosΦ) | Voltage<br>(V) | Limits of variation<br>(%) | Test result (%) |       |       |
|-----------------|------------------------|----------------|----------------------------|-----------------|-------|-------|
|                 |                        |                |                            | №1              | №2    | №3    |
| $I_{min}$       | 1.0                    | 207            | 0.5                        | 0.01            | -0.02 | -0.02 |
| $I_n$           | 1.0                    | 207            | 0.5                        | -0.03           | 0.04  | -0.03 |
| $I_{max}$       | 1.0                    | 207            | 0.5                        | 0.05            | -0.13 | -0.01 |
| $0.1I_n$        | 0.5L                   | 207            | 1.0                        | 0.03            | -0.02 | 0.00  |
| $I_n$           | 0.5L                   | 207            | 1.0                        | 0.02            | -0.01 | -0.02 |
| $I_{max}$       | 0.5L                   | 207            | 1.0                        | -0.05           | -0.17 | 0.01  |
| $I_{min}$       | 1.0                    | 253            | 0.5                        | 0.02            | -0.02 | -0.02 |
| $I_n$           | 1.0                    | 253            | 0.5                        | -0.06           | 0.09  | -0.02 |
| $I_{max}$       | 1.0                    | 253            | 0.5                        | 0.07            | -0.16 | 0.01  |
| $0.1I_n$        | 0.5L                   | 253            | 1.0                        | 0.03            | 0.01  | -0.01 |
| $I_n$           | 0.5L                   | 253            | 1.0                        | 0.02            | 0.05  | 0.01  |
| $I_{max}$       | 0.5L                   | 253            | 1.0                        | 0.03            | -0.11 | -0.01 |
| $I_{min}$       | 1.0                    | 184            | 1.5                        | 0.02            | 0.02  | 0.01  |
| $I_n$           | 1.0                    | 184            | 1.5                        | -0.01           | 0.01  | 0.06  |
| $I_{max}$       | 1.0                    | 184            | 1.5                        | 0.02            | -0.10 | -0.07 |
| $0.1I_n$        | 0.5L                   | 184            | 3.0                        | 0.03            | 0.03  | 0.00  |
| $I_n$           | 0.5L                   | 184            | 3.0                        | 0.02            | 0.04  | 0.02  |
| $I_{max}$       | 0.5L                   | 184            | 3.0                        | 0.04            | -0.13 | 0.01  |
| $I_{min}$       | 1.0                    | 264.5          | 1.5                        | 0.04            | 0.03  | 0.03  |
| $I_n$           | 1.0                    | 264.5          | 1.5                        | 0.01            | -0.02 | 0.03  |
| $I_{max}$       | 1.0                    | 264.5          | 1.5                        | 0.05            | -0.15 | 0.02  |
| $0.1I_n$        | 0.5L                   | 264.5          | 3.0                        | 0.05            | 0.05  | 0.03  |
| $I_n$           | 0.5L                   | 264.5          | 3.0                        | 0.04            | 0.03  | 0.04  |
| $I_{max}$       | 0.5L                   | 264.5          | 3.0                        | 0.08            | -0.10 | 0.06  |
| $I_{min}$       | 1.0                    | <184           | -100 ~ 10                  | Pass            | Pass  | Pass  |
| $I_n$           | 1.0                    | <184           | -100 ~ 10                  | Pass            | Pass  | Pass  |
| $I_{max}$       | 1.0                    | <184           | -100 ~ 10                  | Pass            | Pass  | Pass  |
| $0.1I_n$        | 0.5L                   | <184           | -100 ~ 10                  | Pass            | Pass  | Pass  |
| $I_n$           | 0.5L                   | <184           | -100 ~ 10                  | Pass            | Pass  | Pass  |
| $I_{max}$       | 0.5L                   | <184           | -100 ~ 10                  | Pass            | Pass  | Pass  |
| Test conclusion |                        |                |                            | Pass            | Pass  | Pass  |

## 6.7 Ambient temperature variation

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: The determination of the mean temperature coefficient for a given temperature shall be made over a temperature range 10K above and 10 K below that temperature, but in no case shall the temperature be outside the specified operating temperature range.
3. Test equipment: High and low temperature test chambers PL-2GT(920218) and Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Current: 5(80)A Reference voltage :230V

| Current         | Power factor<br>(cosΦ) | Temperature<br>(K) | Limits of mean<br>temperature coefficient<br>(%/K) | Test result (%/K) |        |        |
|-----------------|------------------------|--------------------|----------------------------------------------------|-------------------|--------|--------|
|                 |                        |                    |                                                    | №1                | №2     | №3     |
| $I_{min}$       | 1.0                    | -25°C~ -5°C        | 0.05                                               | 0.004             | 0.006  | 0.004  |
| $I_n$           | 1.0                    |                    | 0.05                                               | -0.008            | -0.006 | 0.002  |
| $I_{max}$       | 1.0                    |                    | 0.05                                               | -0.004            | 0.007  | 0.006  |
| $0.2I_n$        | 0.5L                   |                    | 0.07                                               | -0.008            | -0.010 | 0.010  |
| $I_n$           | 0.5L                   |                    | 0.07                                               | 0.010             | 0.004  | -0.012 |
| $I_{max}$       | 0.5L                   |                    | 0.07                                               | 0.018             | 0.008  | -0.016 |
| $I_{min}$       | 1.0                    | -5°C~15°C          | 0.05                                               | -0.006            | 0.006  | 0.007  |
| $I_n$           | 1.0                    |                    | 0.05                                               | -0.010            | -0.007 | 0.009  |
| $I_{max}$       | 1.0                    |                    | 0.05                                               | -0.006            | -0.012 | -0.005 |
| $0.2I_n$        | 0.5L                   |                    | 0.07                                               | 0.006             | 0.013  | -0.006 |
| $I_n$           | 0.5L                   |                    | 0.07                                               | 0.005             | -0.004 | -0.006 |
| $I_{max}$       | 0.5L                   |                    | 0.07                                               | -0.012            | -0.005 | -0.008 |
| $I_{min}$       | 1.0                    | 15°C~35°C          | 0.05                                               | 0.006             | 0.011  | 0.006  |
| $I_n$           | 1.0                    |                    | 0.05                                               | 0.005             | -0.008 | -0.004 |
| $I_{max}$       | 1.0                    |                    | 0.05                                               | -0.008            | 0.007  | -0.006 |
| $I_{min}$       | 0.5L                   |                    | 0.07                                               | 0.008             | 0.014  | -0.004 |
| $I_n$           | 0.5L                   |                    | 0.07                                               | 0.002             | -0.008 | -0.006 |
| $I_{max}$       | 0.5L                   |                    | 0.07                                               | -0.004            | -0.005 | -0.004 |
| $I_{min}$       | 1.0                    | 35°C~55°C          | 0.05                                               | -0.006            | 0.007  | 0.007  |
| $I_n$           | 1.0                    |                    | 0.05                                               | 0.010             | 0.008  | 0.011  |
| $I_{max}$       | 1.0                    |                    | 0.05                                               | 0.012             | -0.004 | 0.006  |
| $0.2I_n$        | 0.5L                   |                    | 0.07                                               | -0.005            | 0.008  | -0.006 |
| $I_n$           | 0.5L                   |                    | 0.07                                               | -0.007            | -0.009 | -0.007 |
| $I_{max}$       | 0.5L                   |                    | 0.07                                               | -0.003            | -0.005 | -0.004 |
| Test conclusion |                        |                    |                                                    | Pass              | Pass   | Pass   |



## 6.8 Interruption of phase voltage (Not applicable)

## 6.9 Frequency variation

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: Test the variation in percentage errors when frequency variation at  $\pm 2\%$ .
3. Test equipment: Meter calibration device ST9001D5 (7131019)
4. Test result:

 Reference frequency: 50Hz Current: 5(80)A Reference voltage: 230V

| Current         | Power factor<br>( $\cos\Phi$ ) | Frequency<br>(Hz) | Limits of variation<br>(%) | Test result (%) |       |       |
|-----------------|--------------------------------|-------------------|----------------------------|-----------------|-------|-------|
|                 |                                |                   |                            | №1              | №2    | №3    |
| $I_{min}$       | 1.0                            | 49                | 0.5                        | 0.02            | 0.00  | 0.00  |
| $I_n$           | 1.0                            | 49                | 0.5                        | -0.02           | 0.00  | 0.00  |
| $I_{max}$       | 1.0                            | 49                | 0.5                        | 0.05            | -0.11 | -0.08 |
| $0.1I_n$        | 0.5L                           | 49                | 0.7                        | 0.04            | -0.08 | -0.07 |
| $I_n$           | 0.5L                           | 49                | 0.7                        | 0.03            | -0.06 | 0.00  |
| $I_{max}$       | 0.5L                           | 49                | 0.7                        | -0.03           | -0.01 | -0.02 |
| $I_{min}$       | 1.0                            | 51                | 0.5                        | 0.02            | 0.00  | 0.03  |
| $I_n$           | 1.0                            | 51                | 0.5                        | -0.01           | 0.00  | 0.00  |
| $I_{max}$       | 1.0                            | 51                | 0.5                        | 0.04            | -0.09 | 0.00  |
| $0.1I_n$        | 0.5L                           | 51                | 0.7                        | -0.02           | -0.01 | 0.00  |
| $I_n$           | 0.5L                           | 51                | 0.7                        | 0.02            | -0.01 | 0.03  |
| $I_{max}$       | 0.5L                           | 51                | 0.7                        | -0.03           | -0.01 | 0.04  |
| Test conclusion |                                |                   |                            | Pass            | Pass  | Pass  |

## 6.10 Reversed phase sequence (Not applicable)

## 6.11 Auxiliary voltage variation (Not applicable)

## 6.12 Operation of auxiliary devices (Not applicable)

## 6.13 Short time overcurrents

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: After the application of the short- time over current, the meter shall perform correctly when back to its initial working.
3. Test equipment: Pulse current test bench XTS-12D(03017) and Meter calibration device ST9001D5 (7131019)
4. Test result:

 Reference frequency: 50Hz Current: 5(80)A Reference voltage: 230V

| Current         | Power factor<br>( $\cos\Phi$ ) | Limits of variation<br>(%) | Test result (%) |      |      |
|-----------------|--------------------------------|----------------------------|-----------------|------|------|
|                 |                                |                            | №1              | №2   | №3   |
| $I_n$           | 1.0                            | 1.5                        | 0.02            | 0.02 | 0.04 |
| Test conclusion |                                |                            | Pass            | Pass | Pass |



## 6.14 Self-heating

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test Method: After the voltage circuits have been energized at reference voltage for at least 2h for class 1, without any current in the current circuits, the maximum current shall be applied to the current circuits, the meter error shall be measured at unity power factor immediately after the current is applied.
3. Test Equipment: Meter calibration device ST9001D5 (7131019)
4. Test Result:

 Reference frequency: 50Hz Current: 80A Reference voltage: 230V

| Current         | Power factor<br>(cosΦ) | Limits of variation<br>(%) | Maximum change (%) |      |      |
|-----------------|------------------------|----------------------------|--------------------|------|------|
|                 |                        |                            | №1                 | №2   | №3   |
| $I_{max}$       | 1.0                    | 0.7                        | 0.07               | 0.09 | 0.08 |
| $I_{max}$       | 0.5L                   | 1.0                        | 0.07               | 0.07 | 0.09 |
| Test conclusion |                        |                            | Pass               | Pass | Pass |

## 6.15 Fast load current variations

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test Method: IEC62053-21:2020 IEC62052-11:2020
3. Test Equipment: Meter calibration device ST9001D5 (7131019)
4. Test Result:

 Reference frequency: 50Hz Current: 5(80)A Reference voltage: 230V

| Current         | Requirement                                       | Power factor<br>(cosΦ) | Limits of variation<br>(%) | Maximum change (%) |      |      |
|-----------------|---------------------------------------------------|------------------------|----------------------------|--------------------|------|------|
|                 |                                                   |                        |                            | №1                 | №2   | №3   |
| $I_n$           | $t_{on} = 10\text{ s}$<br>$t_{off} = 10\text{ s}$ | 1.0                    | 2.0                        | 0.09               | 0.05 | 0.08 |
| $I_n$           | $t_{on} = 5\text{ s}$<br>$t_{off} = 5\text{ s}$   | 1.0                    | 2.0                        | 0.11               | 0.04 | 0.09 |
| $I_n$           | $t_{on} = 5\text{ s}$<br>$t_{off} = 0.5\text{ s}$ | 1.0                    | 2.0                        | 0.07               | 0.09 | 0.10 |
| Test conclusion |                                                   |                        |                            | Pass               | Pass | Pass |

## 6.16 Earth fault (Not applicable)

## 7 Tests of the effect of the climatic environments

### 7.1 Dry heat test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: After dry heat test, the meter shall show no damage or change of the information and shall operate correctly.
3. Test equipment: High and low temperature test chambers PL-2G(920218)and Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Current: 5(80)A Reference voltage :230V

| Current         | Power factor<br>(cosΦ) | Percentage error limits<br>(%) | Test result (%) |             |             |
|-----------------|------------------------|--------------------------------|-----------------|-------------|-------------|
|                 |                        |                                | №1              | №2          | №3          |
| $I_n$           | 1.0                    | ±0.5                           | 0.10            | 0.05        | 0.12        |
| Test conclusion |                        |                                | <b>Pass</b>     | <b>Pass</b> | <b>Pass</b> |

### 7.2 Cold test

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: After cold test, the meter shall show no damage or change of the information and shall operate correctly.
3. Test equipment: High and low temperature test chambers PL-2G(920218)and Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Current: 5(80)A Reference voltage :230V

| Current         | Power factor<br>(cosΦ) | Percentage error limits<br>(%) | Test result (%) |             |             |
|-----------------|------------------------|--------------------------------|-----------------|-------------|-------------|
|                 |                        |                                | №1              | №2          | №3          |
| $I_n$           | 1.0                    | ±0.5                           | 0.05            | 0.08        | 0.11        |
| Test conclusion |                        |                                | <b>Pass</b>     | <b>Pass</b> | <b>Pass</b> |

### 7.3 Damp heat, cyclic test

1. Requirement: IEC62053-21:2003 IEC62052-11: 2003
2. Test method: After damp heat cyclic test, the meter shall show no damage or change of the information and shall operate correctly.
3. Test equipment: Damp heat cyclic test chambers SETH-101U/P(634003) and Meter calibration device ST9001D5 (7131019)
4. Test result:

Reference frequency :50Hz Current: 5(80)A Reference voltage :230V

| Current         | Power factor<br>(cosΦ) | Percentage error limits<br>(%) | Test result (%) |             |             |
|-----------------|------------------------|--------------------------------|-----------------|-------------|-------------|
|                 |                        |                                | №1              | №2          | №3          |
| $I_n$           | 1.0                    | ±0.5                           | 0.14            | 0.15        | 0.08        |
| Test conclusion |                        |                                | <b>Pass</b>     | <b>Pass</b> | <b>Pass</b> |

## 7.4 Protection against solar radiation

1. Requirement: IEC62053-21:2020 IEC62052-11:2020
2. Test method: IEC62053-21:2020 IEC62052-11:2020
3. Test equipment: Xenon climate test chamber TET080D (030117)
4. Test result:

| Test condition                                               | Requirement                                                                                                                         | Test result |      |      |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------|------|------|
|                                                              |                                                                                                                                     | №1          | №2   | №3   |
| 8h irradiation and16h darkness<br>3cycles<br>Temperature:55℃ | The appearance and, in particular, the legibility of markings shall not be altered.<br>The function of meter shall not be impaired. | Pass        | Pass | Pass |
| Test conclusion                                              |                                                                                                                                     | Pass        | Pass | Pass |

## 7.5 Durability (No requirement)

**--END OF REPORT--**



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